

Analysis of the Diabetic Foot Ulcer Image Datasets

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Abstract

Coinciding with the ever-ensuing artificial intelligence revolution, deep learning architectures present as a valuable tool in aiding diabetic foot ulcer classification. However, there is a realistic concern regarding the data quality of the datasets utilised for this research area. This thesis employs a systematic methodology in removing near duplicate images in the DFUC2021 dataset using various hashing methods in combination with Hamming distance for a measure of similarity, to produce novel datasets generated from the original. The quality of these datasets was subsequently examined on common deep learning architectures VGG-19 and ResNeXt-50, as well as a modified HRNet-W32 implemented with Sharpness-Aware Minimization (SAM) for minimising the sensitivity of the loss function and enhancing model robustness.

While computational resources posed as a significant limitation for this research, the results of the investigation conclude that removing duplicate images up to a point (95% similarity threshold) improves model performance and therefore dataset quality, but doing so beyond this threshold has counteractive effects due to the inherent limitations of hashing (hash collision, birthday paradox). A more nuanced approach to selecting similarity thresholds, along with a comprehensive quantitative analysis of the various hash algorithms would expand on this research further and improve research outcomes.

Declaration

No part of this project has been submitted in support of an application for any other degree or qualification at this or any other institute of learning. Apart from those parts of the project containing citations to the work of others, this project is my own unaided work.

This work has been carried out in accordance with the Manchester Metropolitan University research ethics procedures, and has received ethical approval (approval number 68872)

Signed: Omari Ricketts

Date: 27-09-2024

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Abbreviations

DFU	Diabetic Foot Ulcer
CNN	Convolutional Neural Network
DFUC	Diabetic Foot Ulcer Challenge
ANN	Artificial Neural Network
RF	Random Forest
AI	Artificial Intelligence
K-NN	K-Nearest-Neighbours
SVM	Support Vector Machines
HSV	Hue, Saturation, Value
WHO	World Health Organisation
LBR	Local Binary Representation
SINBAD	Site, Ischemia, Neuropathy, Bacterial infection, Area, and Depth
PEDIS	Perfusion, Extent, Depth, Infection, and Sensation
A-Hash	Average Hash
P-Hash	Perceptual Hash
G-Hash	Gradient Hash
W-Hash	Wavelet Hash

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1. Introduction

1.1 Diabetes

Commonly referred to as diabetes, diabetes mellitus is a chronic disease characterised by high blood sugar levels (hyperglycaemia), which can lead to severe complications such as cardiovascular diseases, kidney failure, blindness, and lower limb amputation (Hippisley-Cox, 2016). It is well established that insulin deficiency - whether absolute or relative - underlies the biochemical disruptions that contribute to these complications (American Diabetes Association, 2009). There are two primary forms of diabetes. Type 1 causes a complete lack of insulin as a result of the immune system targeting and eliminating the pancreatic beta cells that produce insulin. On average, people with Type 1 diabetes experience a life expectancy reduction of over 20 years (Diabetes, 2019). Type 2 diabetes, in contrast, is marked by insulin resistance, meaning the body's cells do not respond properly to insulin. Eventually, the pancreas becomes unable to produce enough insulin to overcome this resistance leading to a relative deficiency of insulin (American Diabetes Association, 2009). Individuals with Type 2 diabetes, on average, experience a reduction in life expectancy of up to 10 years.

The World Health Organization's global report on diabetes reveals a significant rise in the number of individuals affected by the condition, escalating from 108 million in 1980 to 422 million by 2014. This represents an increase in prevalence from 4.7% to 8.5% of the adult population reflecting the growing trend in unhealthy eating and the consumption of fast food. Currently, it is estimated that 537 million people are living with diabetes and its complications and by the end of 2045, it is projected that approximately 1 in 8 adults will be diabetic, corresponding to approximately 783 million people (International Diabetes Federation, 2021).

It is significant to highlight that approximately 80% of these individuals will be from developing countries, while only about 20% will be from developed nations. This disparity is attributed to lesser awareness and limited healthcare facilities in developing regions (Boulton, 2005). A patient with diabetes requires ongoing expensive medication, frequent check-ups with their physician and sanitary personal care. Patients and their families are consequently severely financially burdened, especially in underdeveloped countries where the expense of treating the disease can equal between five to seven years' worth of income (Cavanagh, 2012). With an estimated 46% of diabetic individuals going undiagnosed, there is an urgent need to address the rising worldwide diabetes pandemic with increased awareness and more easily accessible treatment alternatives.

1.2 Diabetic Foot Ulcers

Diabetic foot ulcers (DFUs) are a severe complication of diabetes impacting approximately 15% of individuals with the condition at some point in their lives. Defined as full-thickness skin lesions penetrating through the dermis located below the ankle in diabetic patients irrespective of their duration, this definition also encompasses skin necrosis and gangrene as ulcers but does not include lesions such as blisters mycosis and, consequently, the term "ulcer" can be ambiguous (Schaper, 2004).

A major contributing factor in the onset of DFUs is neuropathy, a disorder characterised by damage to the peripheral nerves. Diabetic patients can experience different types of neuropathy, but symmetrical distal polyneuropathy - which affects sensation, muscle control, and automatic functions - is especially important in the formation of DFUs (Laing, 1998). Motor involvement leads to muscle weakness and foot deformities that alter how weight is distributed across the feet, increasing the risk of ulceration. Sensory involvement impairs the ability to feel pain, temperature, and touch, allowing minor issues such as blisters to go unnoticed and worsen, and autonomic involvement affects the nerves that control involuntary functions such as sweating. Reduced sweating causes dry, cracked skin that is more prone to injury and infection (Laing, 1998). These factors all increase susceptibility to diabetic foot ulcers and create conditions that allow infections to develop and thrive.

In individuals with diabetes, high blood glucose levels result in the production of advanced glycation end-products which damage the endothelial cells lining the blood vessels and promote the accumulation of fatty deposits in the arteries (Al Suwaidi, 2007). Additionally, diabetes hampers the production of nitric oxide, a vital molecule for the dilation of blood vessels, and increases blood clotting tendencies, further obstructing blood flow (Al Suwaidi, 2007). Consequently, diabetes causes damage and the narrowing of both large and small blood vessels resulting in ischaemia - a condition where reduced blood flow deprives tissues of oxygen and nutrients required for function, potentially causing tissue damage and necrosis. Amputation may be necessary in severe cases of ischaemia in order to stop the spread of necrotic tissue, a consequence that is notoriously associated with diabetes.

In the context of diabetic foot ulcers, ischaemia and infection are closely intertwined. As ischaemia impairs blood flow, this compromised blood supply hinders the immune system's ability to reach and combat infections effectively. Consequently, ischaemic tissues are more susceptible to infections due to their weakened state and reduced ability to heal. When an infection sets in, it can exacerbate ischaemia by causing further inflammation and swelling, which can compress blood vessels and further restrict blood flow. This creates a vicious cycle where ischaemia and infection continuously aggravate each other (Wernick, 2023).

There is a significant death rate from diabetic foot ulcers with around 50% of cases worldwide dying from DFUs within five years (Brennan, 2016). Early detection and intervention are crucial to minimising problems, and effective treatment and preventative measures, such as appropriate footwear, glycaemic control, regular foot inspections, and correct foot care education can dramatically minimise the prevalence of DFUs (Swaminathan, 2024).

2. Current Methods and Challenges In Detecting Diabetic Foot Ulcers

2.1 Clinical Methods

Early detection and intervention are critical components in managing diabetes and its consequences (Herman, 2015). Diabetic foot ulcers manifest differently from person to person, making it a complex and heterogeneous condition. They often resemble other skin conditions, such as fungal infections, and may not exhibit typical signs of infection, especially in patients with poor circulation or neuropathy.

Furthermore, they are multifactorial, involving neuropathy, ischaemia, infection, and other factors. Therefore, a comprehensive diagnostic approach is necessary to identify important risk factors and make an informed, treatment-focused assessment of DFUs. Podiatrists and diabetes experts usually do assessments in foot clinics and hospitals. These evaluations include a thorough analysis of the patient's medical history, foot health, and wound status in addition to a methodical and thorough physical examination (Neves, 2012).

In addition to early detection, other core principles in the effective management of DFUs include culturing the bacteria responsible, treating aggressively with antibiotic therapy, and evaluating the need for debridement and surgery (Edmonds, 2009). No single system can fully encompass the nuances of DFUs leading to the development of various classification systems. Table 1 highlights common classification systems.

Classification System	Description	Grades/Stages	Reference
Wagner Ulcer Classification System	Categorizes ulcers based on depth and tissue involvement.	Grade 0: No open lesions Grade 1: Superficial ulcers Grade 2: Deep ulcers without abscess or osteomyelitis Grade 3: Deep ulcers with abscess or osteomyelitis Grade 4: Gangrene of part of the foot Grade 5: Extensive gangrene	Wagner, 1987
University of Texas Wound Classification System	Grades ulcers from pre- or post-ulcerative lesions to ulcers reaching bone or joint, with stages for infection and ischemia.	Grade 0: Pre- or post-ulcerative lesions Grade 1: Superficial ulcer Grade 2: Deep ulcer without bone involvement Grade 3: Ulcer reaching bone or joint Stages A-D: Stages infection and ischemia	Lavery, 1996
SINBAD Classification System	Uses a cumulative score based on six factors to predict ulcer severity and outcomes.	Site, Ischemia, Neuropathy, Bacterial Infection, Area, Depth	Ince, 2008
PEDIS Classification System	Standardizes assessment of diabetic foot infections using five criteria	Perfusion, Extent, Depth, Infection, Sensation	Mendes, 2012

Table 1 – Common Classification Systems

While the Wagner, University of Texas, SINBAD, and PEDIS classification systems for DFUs all share key similarities - such as a graded approach to ulcer severity, assessment of infection presence and severity, and evaluation of blood flow and ischaemia - studies assessing the accuracy and validity of various diagnostic tests for infection in diabetic foot disease remain limited (Cassidy, 2021). Ideally, a single, unified classification system should be established for general clinical practice to streamline diagnosis and treatment.

2.2 Digital Health Technologies

Digital health tools give healthcare providers a more comprehensive view of patient health by granting access to detailed patient data and empowering individuals to take a more active role in managing their own care. These tools encompass a range of technologies, including mobile health information technology, wearable devices, telehealth, telemedicine, and mobile internet devices (Al Kuwaiti, 2023). Healthcare can benefit from these technological integrations in for remote management of chronic disorders outside of typical care facilities, early diagnosis of dangerous diseases, and prevention. Glucose pumps, continuous glucose monitors, and health-tracking applications, for example, let patients watch their blood sugar levels, control their food and exercise, and get real-time health information. These solutions encourage medication adherence, which enhances overall disease management and improves patient outcomes by providing patients with regular reminders and ensuring they comply with their recommended treatment regimens.

Advanced automated telemedicine systems have emerged as the most economical solution for remote detection of diabetic foot ulcers, offering a minimally invasive and highly accessible approach to care (Al Kuwaiti, 2023). Using key technologies like digital photography for high-resolution images, optical scanners for precise wound depth and area measurement, three-dimensional wound imaging for thorough assessments, and video conferencing for real-time consultations, these systems make it easier to provide health services whenever and wherever necessary. Nonetheless, endocrinologists are still required to complete patient assessments, which inherently carry the risk of human error. The absence of a physical check during video conferences increases this risk further. While these systems show promise, intelligent solutions that are capable of remotely and automatically identifying DFU pathologies are desperately needed. Van Netten (2017) found that clinicians often achieve low validity and reliability when remotely assessing DFUs from foot images. This implies that for even accomplished podiatrists, diagnosing DFUs solely based on images remains extremely challenging.

2.3 Artificial Intelligence and Healthcare

Artificial intelligence (AI) has emerged as a crucial tool for big data applications in healthcare, significantly enhancing clinical services through advanced analytics and the minimisation of human error. By leveraging big data, AI and machine learning algorithms improve electronic health records and accelerate their utilization in health systems. Diverse information from various digital health technologies can be integrated using innovative AI technologies, allowing for precise insights into patient care drawn from extensive research data, user experiences, combined data sources and large datasets (Al Kuwaiti, 2023).

Moreover, AI has shown encouraging outcomes in the early diagnosis of diseases such as skin and breast cancer, eye disorders, and pneumonia through various body imaging modalities (Al Kuwaiti, 2023). Most importantly for the purposes of this investigation, AI holds significant potential for diabetes foot ulcer classification, providing promising paths towards better diagnosis and treatment.

2.4 Ethical Considerations

While the potential for AI to address urgent challenges is exciting, concerns persist. Due to the natural learning curve and quick development of AI technology, errors and unintentional consequences are unavoidable. Safety and ethical issues related to AI must be prioritised in order to responsibly manage these effects. To ensure that AI technology is developed in accordance with moral standards and to protect the well-being of affected populations, data scientists, product managers, data engineers, and domain specialists must work together in concert (The Alan Turing Institute, 2019). The emergence of AI ethics stems from the imperative to mitigate potential harm caused by AI systems, both to individuals and society at large.

Given that AI lacks personal experience, it may appear to be immune to bias from an outside perspective. However, machines learn only from the data they are provided, and incomplete, erroneous, or biased data will always impact AI results. These biases have the power to skew outcomes, producing unfair and inaccurate judgements. The key takeaway is that AI can only be as objective as the training data provided, highlighting the need to identify and address prejudices to ensure AI's ethical application (Ironhack, 2023).

Training AI models for DFU classification involves several nuanced considerations, particularly concerning the data used. These datasets can inadvertently contain biases reflecting systemic inequalities present within the healthcare system. If specific groups are underrepresented in the data, the model may not perform as well for those groups. For example, if data from rural areas or minority populations are sparse, the model might be less accurate for individuals from these groups due to a lack of sufficient training data. Furthermore, the model can mirror these disparities if certain demographics receive less accurate diagnoses. Given that roughly half of the individuals with diabetes are unaware of their condition and many belong to minority populations, it is crucial to recognize that these models may not perform as well for these marginalized groups.

The World Health Organization (WHO) has introduced stringent guidelines crucial for ensuring ethical compliance in developing and deploying AI technologies. These guidelines emphasize several vital principles: Promoting human well-being, safety, and the public interest; protecting human autonomy; ensuring inclusiveness and equity; ensuring transparency, explainability, and intelligibility; fostering responsibility and accountability; and promoting AI systems that are responsive and sustainable. Collectively, these principles aim to establish a robust ethical framework that safeguards both individuals and society in the context of emerging AI technologies (World Health Organisation, 2021).

2.5 Machine Learning and Diabetes

Machine learning-based applications therefore serve as a crucial tool in supporting overburdened healthcare facilities and promoting consistent best practices. These facilitate the automation of labor-intensive tasks and offer decision support directly at the point of care.

High efficacy in diabetes classification can be achieved using traditional machine-learning technologies. Khan et al. (2024) demonstrated this by applying various models, including Artificial Neural Networks (ANN) and Random Forests (RF), to clinical data from the PIMA and Early Diabetes Risk datasets. The study assessed these models along with different feature selection methods. ANN achieved an accuracy of 99.35% on the PIMA dataset, while RF attained 99.36% accuracy on the Early Diabetes Risk dataset. Despite highlighting the effectiveness of machine learning algorithms in predicting diabetes, the requirement for clinical data poses a challenge, particularly for underserved populations with limited access to a qualified healthcare provider.

3. Deep Learning

3.1 Convoluted Neural Networks

Conventional computer vision and machine learning algorithms encountered considerable limitations in handling large amounts of image data, creating intricate data representations with several levels of abstraction, and frequently necessitating meticulous manual adjustment for each input image (O'Mahony, 2019) (Karypidis, 2022).

The 2012 ImageNet Large Scale Visual Recognition Challenge demonstrated the potential of Convoluted Neural Networks (CNNs) for visual recognition tasks, where AlexNet's remarkable performance decisively surpassed competition, solidifying CNNs' reputation as the pioneering deep learning architecture (Krizhevsky, 2012). Significant progress has since been made in various computer vision disciplines, such as object identification, segmentation, and classification. The success of Convolutional Neural Networks is largely due to their architectural structure and functional mechanisms. In image processing tasks, a CNN generally processes a 3-dimensional tensor that represents an image, characterized by H rows, W columns, and three colour channels: red, green, and blue (RGB) (Tensorflow, n.d.). This tensor undergoes a series of processing steps through various layers, with each layer performing a specific function on the image. Convolutional layers apply kernels that slide over the input data, scanning the tensorized image in small segments, and perform element-wise dot products to assign weights to specific areas of the image (Motamedi, 2016).

These weights are combined with a bias value across the entire image to produce a weighted sum, with the product being a feature map for each kernel (Feddewar, 2022). Non-linearity is introduced via activation functions like ReLu (Poudel, 2021), which aids in the propagation of information through the subsequent layers, allowing for the identification of complex semantic features. Advancing through each layer, the data undergoes transformations driven by weights, biases, and activation functions, allowing the original input tensor to evolve into more abstract and meaningful representations. Pooling layers minimise computational effort and spatial dimensions while keeping crucial image details in order to down-sample the resulting feature maps (Yamashita, 2018). Data flows via pooling and convolutional layers before arriving at the output/fully connected layer, which is where high-level features are integrated. The output scores from several categories are normalised using a final activation function, such as the SoftMax function for multi-class classification, and are then translated into probabilities (Hinton, 2012). The most likely categorisation for the input is shown by the final prediction, which matches the category with the highest probability. An epoch ends when a full iteration over the dataset is completed, at which point the backpropagation algorithm is employed. In order to minimise prediction error, this computes gradients of the loss between predicted and ground truth with

weights and biases adjusted. The new epoch is started after the weights are adjusted in the direction that yields the highest accuracy (Zhou, 2018). A significant challenge lies in directly interpreting the meaning of the weights of the kernels and the significance of the feature maps associated with these weights, often resulting in CNNs being referred to as black boxes (Liang, 2020). Gaining deeper insights into these components would certainly improve the capacity to fine-tune CNNs and improve resultant performance, but this undertaking is inherently complex.

3.2 Deep Learning and Diabetes - Annual Diabetic Foot Ulcer Challenges (DFUC)

Goyal et al. (2017) introduced the first fully automatic method leveraging a Convolutional Neural Network, DFU-NET, for DFU classification based off images alone. Their research, conducted on a binary dataset comprising 1,459 photos, highlighted the potential of deep learning techniques in classifying diabetic foot ulcers, achieving impressive precision and accuracy scores of 0.867 and 0.907 respectively. Following the demonstrated efficacy of deep learning methods for DFU classification, significant research in diabetic foot ulcer localisation, classification and segmentation has been driven by the annual Diabetic Foot Ulcer Challenge competitions (DFUC, 2020-2024). The inaugural DFUC2020 concentrated on DFU localisation methods for automated detection, and the competition recognised two top-performing teams: Ryo Hachiuma et al. achieved first place with the highest mean average precision of 0.6940, and Manu Goyal et al., the researcher who introduced DFU-NET, achieved attained the highest F1-score of 0.7434 (Yap, 2021b).

DFUC2021, the second challenge in the series, centered on multi-class classification involving four classes: control, infection, ischaemia, and a combination of infection and ischaemia (both). The challenge was won by Adrian Galdran et al., achieving a macro F1-score of 0.6216 (Cassidy, 2021). The third Diabetic Foot Ulcer Challenge (DFUC2022) focused on segmentation tasks using the DFUC2022 dataset, which is the largest available dataset for DFU segmentation, comprising 40,000 images. Although the DFUC competition did not take place in 2023 for unspecified reasons, the DFUC2024 competition is ongoing, continuing with a segmentation focus that aims to build on the success of DFUC2022.

3.3 Further Developments in DFU Classification

Santos et al. (2022a) introduced DFU-VGG, a novel CNN built on the VGG-19 architecture, fine tuned to specifically classify diabetic foot ulcers into four distinct categories. The model was trained on a combined dataset of DFUC2020 and DFUC2021, encompassing a total of 8,250 images (including augmented). By incorporating smaller dense layers and adding batch normalization after each convolutional block, DFU-VGG achieved an accuracy of 93.45% and a Kappa index of 89.24%. In a subsequent study by Santos et al. (2022b), the investigation into ensemble models that combine various architectures - VGG-16, VGG-19, InceptionV3, ResNet50, and DenseNet201 - further enhanced with modifications to fully connected layers, dropout, and batch normalization was performed. Fine-tuning these models on the same dataset resulted in an impressive average accuracy of 95.04% and a Kappa index exceeding 91.85%, building on the progress made in their initial investigation.

Khandakar et al. (2021) evaluated the effectiveness of traditional machine learning techniques in conjunction with CNNs versus shallow convolutional neural networks alone, specifically MobileNetV2, for classifying thermogram images of diabetic foot ulcers. The study utilized a pre-trained CNN model to extract features from DFU images and applied k-means clustering to categorize the images into three levels: mild, moderate, and severe. An AdaBoost-classifier was employed, leveraging a random forest model for feature selection with ten features. MobileNetV2 achieved an impressive F1 score of 95% for classifying thermogram images, while the AdaBoost-classifier, using ten features, attained an F1 score of 97%, outperforming the CNN. The comparison of inference times revealed that the proposed method is also suitable for use as a smartphone application, enabling users to monitor DFU progression at home thus increasing accessibility.

Preiya et al. (2023) presents a method for classifying DFUs using a Pre-Trained Fast Convolutional Neural Network integrated with a Deep Recurrent Neural Network. Trained on DFUC2021, the dataset was pre-processed and segmented with features extracted using a deep recurrent neural network. Normal and abnormal diabetes ranges were identified and it categorised ulcers into five stages, from normal (stage 1) to necrotic (stage 5), with stages 3-5 requiring immediate medical attention. The Pre-Trained Fast Convolutional Neural Network achieved a high classification accuracy of 99.32%. Although this study, as well as the one by Khandakar (2021), do not explicitly classify diabetes status, both effectively distinguish between normal and abnormal diabetes ranges with high accuracy, offering a valuable tool for identifying when medical intervention is necessary.

Despite the demonstrated potential of deep learning and machine learning methods for automating DFU classification, several limitations have been identified in current research methods. They highlight the urgent need for a publicly accessible, multimodal dataset that includes diverse imaging types for diabetic foot patients. Current machine learning models, which can achieve over 99% accuracy based on clinical data, often rely on healthcare accessibility that may be costly or unavailable to some populations. To bridge this gap, it is essential to develop comprehensive datasets that incorporate various imaging modalities, such as thermal infrared (for early ulcer detection), clinical wound images (to monitor ulcer progression), fluorescence (for identifying clinically significant bacteria), and MRI (for detecting infections and Charcot's foot). Collating such datasets and training CNN models on this data would greatly enhance the accuracy of DFU classification on images alone for instances where clinical data is absent, greatly advancing the field and potentially enabling automated systems to replace manual diagnostic methods to establish an internationally recognized diagnostic approach.

4.0 Diabetic Foot Ulcer Datasets for Classification

4.1 DFU Part A

Goyal et al. (2018) introduced the DFU Part A dataset, specifically tailored for classification tasks related to diabetic foot ulcers. The dataset comprises 397 foot images, 292 of which show diabetic foot ulcers and 105 of which show healthy feet. Podiatrists delineated 1,679 skin patches from these images, with 641 patches representing healthy skin and 1,038 patches indicating DFUs. These images were collected over five years from Lancashire Teaching Hospitals, with all necessary ethical approvals and patient consent obtained.

To ensure high image quality, the Nikon AF-S DX Micro NIKKOR 40 mm f/2.8G lens was used, avoiding flash and relying on consistent room lighting. Additionally, a heterogeneous test set was included, featuring 20 abnormal and 32 normal skin patches captured with an iPad using the FootSnap application for algorithm control purposes. Medical professionals provided ground truth labels recorded in XML files, and these labels were used to create manually annotated segmentation masks that highlight the ulcer regions. The masks are binary, with one class representing the ulcer and the other the background.

During their investigation, Goyal et al. (2018) divided the samples into training (1,423 patches), validation (84 patches), and testing (172 patches) sets. To enhance the dataset and increase generalisability of the model, various data augmentation techniques were implemented, increasing the dataset size by 15 to 21,345 training patches and 1,260 validation patches. As a result, DFU-NET achieved respectable performance metrics: Sensitivity of 0.934, F-measure of 0.939, and AUC of 0.962 in binary classification tasks of diabetic foot ulcers.

While these metrics highlight the model's strong capability in distinguishing between ulcerative and non-ulcerative conditions, the binary classification approach does not fully capture the complexity of diabetic foot ulcers, which often manifest in multiple forms beyond a simple binary categorization.

4.2 DFU Part B

Similar to DFU Part A, DFU part B focuses on binary classification tasks, but it investigates explicitly infection-vs-all and ischaemia-vs-all classifications. Two consultant physicians specialising in diabetic foot care provided the ground truth through a visual examination of DFU photographs. In cases of disagreement, the more experienced doctor made the final decision. The dataset consists of 3,332 DFU patches derived from 2918 images, with 235 labelled as ischaemic and 1,431 as non-ischaemic. Additionally, 684 patches were labelled as free of bacterial infection, while 982 showed signs of infection. Amin et al. (2020) obtained respectable F1-scores using an ensemble CNN: 0.902 for the ischaemia-vs-all classification and 0.722 for the infection-vs-all classification.

Despite the proven efficacy and progress made in addressing issues identified in the DFU Part A dataset, challenges persisted with the DFU Part B dataset. Expert labelling remains essential, as the dataset's quality directly affects the analysis's quality. Due to the heterogeneous nature of DFUs, research utilising the dataset have found substantial challenges attributed to the high intraclass dissimilarities within DFU classifications as well as the strong interclass similarities between DFU skin, healthy skin, and other skin lesions. Furthermore, employing a CNN to classify infection or ischaemia can lead to missing the co-occurrence of ischaemia and infection instances due to the dataset's binary classification nature. This poses as a limitation for practical applications where the ultimate goal is to replace manual classification.

4.3 DFUC2021

Cassidy et al. (2021) developed the DFUC2021 dataset, the largest available resource for diabetic foot ulcer pathology for classification, containing photographs collected from patients at Lancashire Teaching Hospitals during clinical visits. Unlike earlier binary classification datasets, DFUC2021 is unique in that it has multi-class classification capabilities. Originally, it was comprised of 1,703 images of ulcers with infection, 152 ulcers with ischaemia, 372 ulcers with both conditions, 1,703 control samples (classification none), and 1,337 unlabelled patches. Two healthcare experts specialising in diabetic wounds and ulcers developed the ground truth labels for the dataset, with annotations based on the presence of ischaemia, infection, both conditions, or neither, validated using patient medical records.

Augmentation techniques were applied to enhance deep learning algorithms, resulting in a total of 15,683 DFU patches. This includes 11,689 labelled patches - 5,955 for training and 5,734 for testing - and 3,994 unlabelled patches intended for training. The resultant class distribution is as follows: "none" for non-infected ulcers (2,552 in training), "infection" for infected ulcers (2,555 in training), "ischaemia" for ischaemia-affected ulcers (227 in training), and "both" for ulcers with both infection and ischaemia (621 in training).

4.4 DFUC2021 Model Performances

The winning model in the DFUC2021 competition employed an effective ensemble strategy, combining two powerful deep Convolutional Neural Networks: BiT-ResNeXt50 and EfficientNet-B3. ResNeXt-50, a variant of the ResNet architecture, adheres to standard Convolutional Neural Network principles with layers for feature extraction but distinguishes itself in this application by using group normalization instead of batch normalization and incorporating weight standardization in its convolutional layers. Additionally, it incorporates ReLU activation functions to introduce non-linearity and applies dropout for regularization. The key innovation of ResNeXt-50 lies in enhancing feature diversity by increasing cardinality within its residual blocks. Grouped convolutions greatly increase the representational power of the network by allowing distinct groups of filters to capture a variety of features (Xie, 2017). The implemented model was pre-trained on ImageNet using the Big Transfer (BiT) approach and with 25.55 million parameters (Galdran 2021).

EfficientNet-B3, part of the EfficientNet family, focuses on balancing efficiency and accuracy. Like other CNNs, it includes convolutional layers, but its unique strength lies in its compound scaling method. EfficientNet simultaneously scales the network's depth (number of layers), width (number of channels), and input resolution (image size), which contrasts with traditional methods that typically scale only one dimension at a time. This ensures that the network grows in a way that optimises performance without incurring unnecessary computational costs. With 12.23 million parameters, EfficientNet-B3 often exhibits superior efficiency in comparison with other architectures with similar performance levels (Galdran 2021).

By integrating predictions from both CNNs, the ensemble-averager model demonstrated a strong performance of CNNs when dealing with a limited number of original data samples. A vital component of the model's success was the use of Sharpness-Aware Minimization (SAM) for optimization, rather than the traditional Stochastic Gradient Descent (SGD), with SAM notably improving the generalization and performance of the models. The winning ensemble model achieved an F1 score of 62.16, an AUC of 88.55, a recall of 65.22, and a precision of 61.40, earning first place in the competition for all metrics except precision.

Bloch et al. (2021) secured second place in DFUC2021 by employing an ensemble model of EfficientNet versions E1, E2, and E3, trained on an extended dataset. Unlike the winning approach, which did not make use of the unlabelled data, this study effectively leveraged such data, incorporating 6,961 pseudo-labelled images along with 5,955 original images and 38,748 synthetically generated images created using the application Pix2PixHD. This resulted in 12,916 images per class, totalling

51,664 images, and expanded the DFUC2021 dataset size by a factor of 8.68. The ensemble model, which generated final predictions via averaging, achieved a macro F1-score of 60.77%, outperforming its baseline counterparts. Model performance demonstrated significant increases in classification performance for rare classes like infection and ischaemia while maintaining or slightly boosting performance for more common classes despite variability in synthetic image quality. This demonstrates how to improve model performance by utilising pseudo-labelling to facilitate semi-supervised learning and by correcting class imbalance.

The technique that claimed third place in DFUC2021 once again employed an ensemble model, this time consisting of fine-tuned Resnet-50, EfficientNets B0-B6, and Resnet-101 (pretrained on ImageNet). To overcome the inherent class imbalance in the dataset, researchers employed a custom activation layer using bias-adjustable SoftMax. They found that because the loss function used class weights, trained networks showed a bias towards the majority classes control and infection cases. A novel custom activation layer was introduced to address this problem by altering the probability skew for each class to change the bias at the inference level (Cassidy, 2021). The researchers managed to achieve a respectable Macro F1-score of 59.29%, with the model architecture proving to be competitive amongst the top performers.

A review of the top-performing models from the DFUC2021 competition revealed the importance of maintaining a high resolution of images for optimal model performance. Deeper architectural models typically beat shallower ones, showing how important complex networks are for managing complicated/nuanced data patterns present in diabetic foot ulcers. The top three entries in the competition employed a combination of different deep architectures, which were then integrated into an ensemble model. This averaging ensemble approach was vital to achieving superior results.

5.0 Duplicates

Delineation is the method of defining/dividing particular regions of an image to produce discrete patches or samples. For DFUs it is used to highlight the regions on the foot showing signs of infection, ischaemia, both or none.

In the datasets discussed, the number of delineated images exceeds the original number of photographed images, indicating that multiple samples have been produced from single images. While segmented, this can introduce the risk of duplicates within the dataset, which is further compounded by the implementation of data augmentation techniques to enhance the dataset further. Despite the intention being to supplement the samples already in the dataset and introduce diversity to improve generalisation of the model on unseen data, overrepresented augmented samples can skew the training process by overemphasising and learning patterns specific to the overrepresented images instead of those common to the classification status. This can result in overfitting, and negatively impact the model's performance due to biased training.

The impact of near-duplicate images on well-known deep learning architectures trained on the DFUC2021 dataset was measured by Dipto (2021). The study examined the effect of visually similar images on model performance and found that, although their inclusion can at first lower accuracy, removing too many can also degrade performance, assumably due to insufficient training data. These results emphasise the impact of using non-overlapping datasets, achieved through duplicate image removal to enable impartial train/test splits, has on model performance. They also underscore the value of diversity of samples in datasets and how a lack thereof can generate biases in model training.

5.1 Near-Duplicate Image Detection

Defining what constitutes a "near-duplicate" can vary depending on the acceptable photometric and geometric variations which determine how closely images resemble each other. Ranging from exact duplicate detection, where no alterations in the image are allowed, to more generalised forms, where the images must depict the same scene but may vary in viewpoint, lighting, rotation, or other conditions. There are numerous studies in the literature exploring the effectiveness of near-duplicate image detection methods, highlighting how performance varies with different image alterations. The choice of an appropriate method for near-duplicate detection is crucial, as it must align with the applications specific requirements. Factors such as tolerance to variations in lighting, viewpoint, and other transformations play a significant role in determining the most suitable feature extraction technique for accurate detection.

5.2 Literature Methods of Duplicate Removal

The paper by Malviya et al. (2016) presents a method for detecting near-duplicate images, specifically targeting copy move forgery which involves duplicating a region within the same image to manipulate its content. It is especially difficult to detect because it leaves no obvious visual clues, but it introduces pixel-level inconsistencies. The proposed method utilizes the HSV(Hue, Saturation, Value) colour space and colour moments to detect tampering. By dividing the image into blocks and analysing colour features and histograms, this approach can identify duplicated regions, even when they undergo transformations such as rotation or scaling, making it robust against such modifications.

The results show significant accuracy in detecting forgeries across various images, including those with multiple duplications. However, despite its strengths, the method has its limitations. It primarily relies on colour features, which may be less effective for detecting forgery in grayscale or highly textured images. Additionally, the block-based feature extraction approach may be less efficient in images with high variability in block content, or where the duplication involves fine-grained details. The method's performance may also deteriorate if the forgery involves techniques like blurring, which alters pixel

statistics significantly. Furthermore, the approach may need optimization for real-time applications, where processing speed and computational resources are critical constraints.

The paper by Singh et al. (2024) presents an architecture for detecting rebroadcast images by integrating a modified CNN with support vector machines (SVM)/k-nearest-neighbours as classifiers. This hybrid model leverages both deep learning and traditional machine learning techniques to achieve high accuracy in detecting near-duplicate images. At the model's core is a CNN designed to extract key features from images using filters that capture patterns such as edges and textures. A distinguishing aspect of this model is its use of Haar wavelet decomposition, which breaks images into horizontal, vertical, and diagonal sub-bands. Diagonal features are particularly effective in distinguishing between original and duplicate images.

For classification, the extracted features are fed into either a support vector machines or k-NN classifier, with SVM consistently outperforming k-NN in differentiating between originals and near-duplicate images. This approach has shown remarkable results, achieving over 92% classification accuracy in rebroadcast image detection across all assessed datasets. Despite its strengths, combining a CNN for feature extraction and SVM or k-NN for classification adds complexity, making the model challenging to implement and fine-tune. Deploying the model in real-time applications or on resource-constrained devices is impeded by the computationally demanding feature extraction process during training.

A thorough analysis of the effects of different image similarity thresholds on the functionality of deep learning classification networks is provided by Dipto et al.'s (2021) study, which sheds light on the optimal conditions for model accuracy. It finds that the maximum validation accuracy with InceptionResNetV2 is achieved at a similarity threshold of 80%, providing useful insight on how to balance the removal of similar pictures. However, the study's reliance on a single image similarity algorithm, dupeGuru, limits its findings and overlooks the potential advantages of other similarity measures. The specific hardware and software configurations used in the study could influence the reproducibility of the results, potentially restricting their applicability to different setups.

Nian et al. (2015) introduces a novel image representation technique called Local Binary Representation (LBR) for efficient and accurate near-duplicate image and video keyframe detection. LBR works by utilizing a compact and robust binary encoding of local feature histograms. To improve detection accuracy, the technique entails optimising sliding window settings, such as block size and step size. Experiments demonstrate that LBR is a useful technique for detecting almost identical pictures and keyframes in films. It achieves good performance in terms of efficiency and execution speed as well as high accuracy and recall. Compact binary encoding is responsible for its computational efficiency. Efficient representation minimises memory usage and storage needs without sacrificing robustness, which makes LBR appropriate for big datasets and real-time applications. However, LBR is prone to false positives, especially when the feature histograms of visually distinct images resemble one another due to the absence of spatial information.

Image hashing is a method for comparing images by converting them into hashes and evaluating the differences between these hashes. Unlike cryptographic hashing, which is highly sensitive to even the most minor changes, image hashing is resilient to minor variations (Bashir, 2019), making it particularly useful for identifying visually similar images that have undergone augmentations. There are several applications for image hashing, including image verification, image search, and near image duplicate detection (Sidharthan, 2017).

White et al. (2020) presents an innovative approach to image authentication by integrating blockchain technology with various image hashing algorithms. The system hashes images using multiple algorithms, normalizes the resulting data, and stores these hashes on a blockchain for easy comparison. To determine the similarity between image hashes, the Hamming distance is employed. The Hamming distance measures the number of differing bits between two binary strings of the same length. For example, if two binary strings differ in three positions their Hamming distance is 3. Therefore, a smaller

Hamming distance indicates a higher similarity between the images as fewer bits are different, while a larger distance signifies more significant differences (Singh, 2024).

The experiment evaluates the system's effectiveness using standard images subjected to different manipulations, such as Gaussian blur, rotation, cropping, and flipping. Results show that the hash-set framework performs well in certain scenarios, particularly with Gaussian blur. The versatility when implementing different hashing algorithms enables robust performance across various image processing situations, offering a reliable image verification and comparison method. Nevertheless, the performance of different hashing algorithms can be inconsistent, with some algorithms struggling more than others when dealing with specific types of alterations. For instance, the system has difficulty handling image rotations, often producing unreliable comparisons and inconsistent results with specific hashing methods. Combining complementary hashing methods that account for the limitations of others would be the best way to implement hashing algorithms for near duplicate image detection.

6. Novel Approach to Dataset

6.1 Preprocessing

NHS Research Ethics Committee approval was acquired for the use of the DFUC2021 dataset. While combining all three datasets was considered, this step was deemed unnecessary as DFUC2021 already includes all the training samples from datasets DFU Part A and DFU Part B.

The test and training datasets were provided as separate zip folders. The training data's ground truth labels were kept in a CSV file, with each label being linked to a specific image by its filename. As a result, matching these ground truth labels to the relevant images in the training dataset was required. To ensure that the input dimensions were appropriate and consistent for processing by the implemented architectures, the resolution of each input image was converted to a square shape with dimensions of 224 x 224 pixels and then transformed to a PyTorch tensor. The images were separated into their respective folders based on ground truth classification.

To address the challenges of insufficient training data and imbalanced class distributions in all of the datasets used for diabetic foot ulcer classification, data augmentation techniques were applied. These techniques aim to enhance the dataset by introducing various modifications to improve performance in visual tasks. Specifically, images were randomly rotated within a 180-degree range and flipped both horizontally and vertically to simulate different orientations (Maharana, 2022). Brightness was adjusted between 70% and 130% of the original value, and shear transformations with a maximum magnitude of 20% were applied to skew the images. Additionally, images were zoomed in and out by up to 20% of their original size (Sánchez-Peralta, 2020). As a result, each class was expanded to include 2600 images per class (10,400) generated from the original DFUC2021 dataset, evening the class distribution with the overall aim of introducing diversity and helping the model learn robust and generalised patterns.

In order to curate novel datasets by identifying and removing near-duplicate images effectively, a combination of methods identified from the literature was employed, striking a balance between efficiency, computational complexity and performance. Specifically, image hashing was used in conjunction with Hamming distance to assess similarity. By converting images into fixed-size hash values, image hashing allows for a comparison of hashes using the Hamming distance which counts the number of bits that differ between two hashes. A similarity threshold is applied to determine if two images are considered as duplicates. This threshold is a predefined value that sets the maximum allowable Hamming distance for two hashes to be classified as similar. For instance, a similarity threshold of 95% means that only images with hashes differing by no more than 5% of their bits are deemed duplicates. By employing hashing in conjunction with Hamming distance to determine similarity threshold, this effectively provides a definition of a near duplicate image and provides an effective solution to evaluate image similarity,

$$\text{Similarity Threshold} = \frac{\text{Hamming Distance}}{\text{Total Number of Bits}} \quad (6.1)$$

A hash size of 16 was selected during this investigation, resulting in a 256-bit hash string for every image. This decision was made by progressively increasing the hash size and evaluating both the number of images removed due to the deduplication and the resultant model performance. The compromise between computational efficiency and detail preservation was effectively achieved with a

hash size of 16. The emphasis in the literature on maintaining a high resolution of the images further supported this decision.

With 2^{256} possible hash combinations (1.15×10^{77}), this method significantly enhances the ability to differentiate between images. Smaller hash sizes, such as 8x8, are faster and easier to compute, but they tend to capture less information increasing the risk of false positives. This is attributed to the coarser representation of images which reduces the ability to detect minor changes. On the other hand, the 256-bit hash used in this study offers a more sensitive representation making it better suited for detecting subtle variations between images without being unduly sensitive to minor ones. The vast number of possible hash combinations decreases the possibility of visually different images being assigned identical/similar hash values, increasing the dependability in differentiating between highly similar images that may have undergone various possible augmentations. The following section details the specific hashing methods employed for the purposes of this investigation, each chosen for its suitability in analysing image similarity.

6.2 Average Hash (A-hash)

The average hash (A-hash) algorithm is a hashing technique that generates compact 256-bit hashes that capture the essential structural details of an image. The image is first scaled down to 256 pixels by creating a 16x16 grid. Then, since the luminance component preserves the important semantic features of an image, each pixel is converted to greyscale, an approach common to hashing algorithms (Zauner, 2010). The average pixel value of the greyscale image is then computed, which is used to create a binary hash where each bit indicates whether the value of a pixel is higher or lower than the average. A pixel's associated bit is set to 1 if its value is higher than the average, if below it is set to 0. This effectively maintains the image's essential elements, condensing its representation into a 256-bit fingerprint. Figure 1 illustrates the identification of an original image using the A-hash method, while Figure 2 shows the corresponding near-duplicate image.



Figure 1 – Original Image Detected via A-hash, Classification Both



Figure 2 – Near Duplicate Image (zoom) Detected via A-hash, Classification Both

6.3 Perceptual Hash (P-hash)

Similar to A-hash, the process starts by converting the image to grayscale, which strips away colour information and focuses on structural features. With a hash size of 16, the grayscale image is then resized to 64x64 pixels (hash size x 4). After condensing the image representation, it undergoes a discrete cosine transform, transforming the image data from the spatial domain (pixel intensity) to the frequency domain (Drmic, 2017). This procedure captures the image's high frequency components, as well as the

low frequency components, which are the most prominent and stable patterns of the image represented in the top left corner of the resulting matrix. Only the top-left 16x16 portion is retained, representing the image's low frequencies which hold the most significant structural information. These low-frequency elements are selected to capture the fundamental structure of the image as they are more resilient to alterations such as scaling, rotation and noise (Drmic, 2017). The median of the values in this 16x16 low-frequency matrix is determined and a binary hash is produced by comparing each value in the matrix to the median. A pixels associated bit is set to 1 if its value is higher than the median, if below it is set to 0. The resulting binary matrix is flattened into a 256-bit array, serving as a unique fingerprint of the image used to compare similarity between images. Figure 3 illustrates the identification of an original image using the P-hash method, while Figure 4 shows the corresponding near-duplicate image.

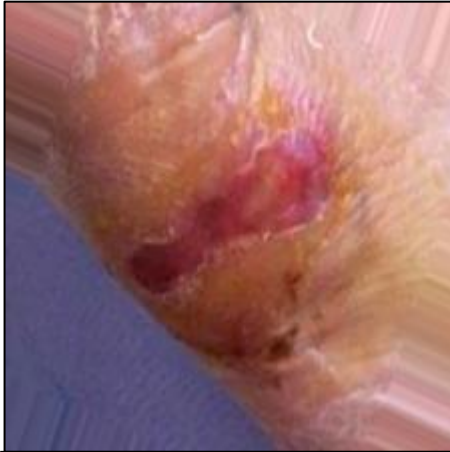


Figure 3 – Original Image Detected via P-hash, Classification None

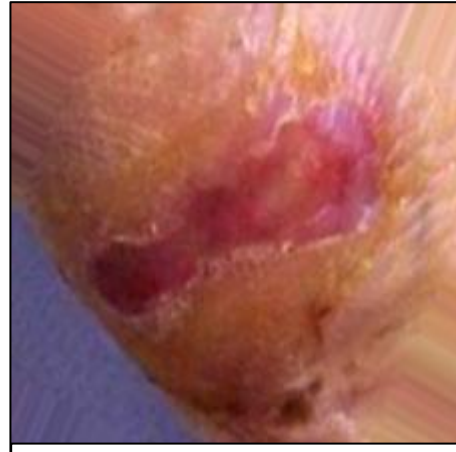


Figure 4– Near Duplicate Image (zoom) Detected via P-hash, Classification None

6.4 Wavelet Hash (W-hash)

Similar to P-hash, W-hash operates in the frequency domain but instead of using the discrete cosine transform it makes use of the discrete wavelet transform, specifically the Haar wavelet transform. This divides the image into approximation and detail coefficients; the former captures the image's broad, low frequency features while the latter captures high frequency details such as sharp edges. The W-hash algorithm eliminates the higher frequencies that represent noise and finer information while retaining the low frequency approximation coefficients which are used to create a binary hash for comparison. W-hash performs particularly well with images featuring fewer intense changes or high-contrast spatial data. (Hamadouche, 2021). Figure 5 illustrates the identification of an original image using the W-hash method, while Figure 6 shows the corresponding near-duplicate image.



Figure 5 – Original Image Detected via W-hash, Classification Both

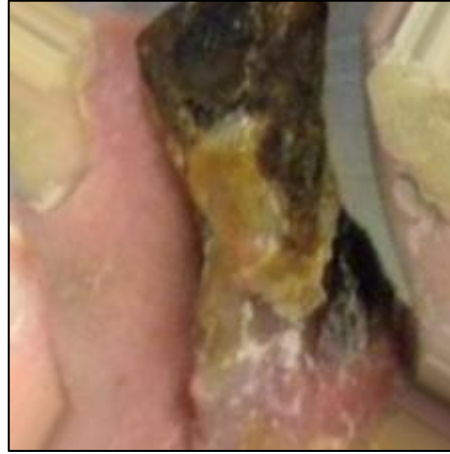


Figure 6– Near Duplicate Image (zoom) Detected via W-hash, Classification Both

6.5 Marr Hildreth Hash

The Marr Hildreth hash algorithm, focusing on edge detection, begins by converting the image to greyscale. Subsequently, a 7x7 kernel is applied to smooth the image using Gaussian blur, reducing noise and detail thus highlighting significant edges. The function takes the blurred greyscale image and the Laplacian operator is applied. The Laplacian operator is a second-order differential operator that calculates the rate of change of the gradient, highlighting areas with rapid intensity fluctuations making it effective at identifying edges as it responds strongly to regions where the image intensity changes abruptly (Wang, 2007). The edge detected image is then resized with dimensions 16x16 pixels, and the median value of the resized image is determined. Each pixel's intensity is compared to this median value to generate a binary array, where a pixels associated bit is set to 1 if more intense than the median, if less it is set to 0. This binary array is flattened into a 256-bit string capturing the image's structural features which is utilized for detecting near-duplicate images. Figure 7 illustrates the identification of an original image using the Marr Hildreth -hash method, while Figure 8 shows the corresponding near-duplicate image.



Figure 7 – Original Image Detected, Classification Infection



Figure 8 – Near Duplicate Image (flip) Detected via Marr Hildreth hash, Classification Infection

6.6 Block Mean Hash

The block mean hash function resizes an image to twice the specified hash size and converts it to greyscale. The average pixel value for each block is then determined by dividing the enlarged image into 2x2 pixel blocks, and the hash is produced by comparing each block's average to the calculated median of these average values; blocks with averages above the median receive a 1, and those with averages below receive a 0. Figure 9 illustrates the identification of an original image using the block mean hash method, while Figure 10 shows the corresponding near-duplicate image.



Figure 9 – Original Image Detected,
Classification Ischaemia

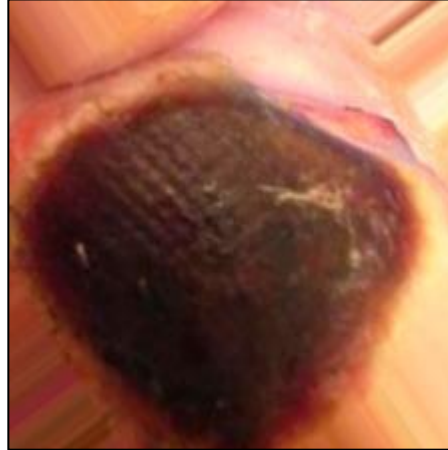


Figure 10 - Near Duplicate Image
(flip) Detected via Block Mean hash,
Classification Ischaemia

6.7 Gradient Hash (G-Hash)

The gradient hash algorithm examines an image's gradients to produce a hash. The image is converted to greyscale, and the gradient magnitudes representing variations in pixel intensity both horizontally and vertically are computed using Sobel filters. Sobel filters calculate changes in pixel values in these directions using specific kernels to emphasize edges. The gradient magnitude image is condensed to a compact 16x16 pixel grid, and the hash is produced by comparing each pixel's gradient magnitude to the calculated median; if greater, the pixel is set to 1, if less it is assigned 0. This process results in a binary hash that provides a compact, distinctive representation of the image's edge information. Figure 11 illustrates the identification of an original image using the gradient hash method, while Figure 12 shows the corresponding near-duplicate image.



Figure 11 – Original Image Detected via Gradient hash, Classification Ischaemia



Figure 12 - Near Duplicate Image (rotate) Detected via Gradient hash, Classification Ischaemia

The different hashing algorithms are complementary strategies that work well in a range of scenarios and for varying objectives. Every technique uses a different strategy to extract the most important information from an image, concentrating on different characteristics such as edge detection, gradient magnitudes, structural information, or frequency components. Because different methods may perform better under different situations, such as noise, transformations, or lighting changes, which are common in real-world datasets, this diversity enables a more rigorous study of picture similarity. This multi-pronged strategy allows for efficient retrieval and comparison of images and also allows for the detection of near-duplicate images that have undergone augmentations that are not viewed as providing additional information.

After applying the various image augmentation techniques discussed, the dataset Aug-DFUC2021 was produced (table 2) which possessed evenly distributed classes. This dataset was then processed using each hashing method outlined using a hash size of 16 and a specific similarity threshold. The deduplication process was conducted both within and across class folders, and adjusting these thresholds allows for a deeper understanding of the similarity between different image hashes and, consequently, the images they represent. Following the deduplication process, the model performances were evaluated on datasets ranging from 100% to 80% similarity thresholds, with increments of 5%. Additionally, the performance of the model trained on the original DFUC2021 dataset, as well as on the augmented dataset without deduplication (Aug-DFUC2021), was analysed. The specific class distributions are detailed in table 2.

Dataset	Ischaemia	Infection	None	Both	Toal
DFUC2021	227	2555	2552	621	5955
Aug-DFUC2021	2600	2600	2600	2600	10400
Aug-DFUC2021 (100%)	2581	2594	2579	2593	10347
Aug-DFUC2021 (95%)	2454	2457	2380	2510	9801
Aug-DFUC2021 (90%)	1640	1714	1481	1870	6705
Aug-DFUC2021 (85%)	1012	1176	951	1376	4515
Aug-DFUC2021 (80%)	431	546	393	665	2035

Table 2 - Count of Class Distribution for Datasets Used During Investigation.

7.0 Performance Metrics

7.1 Macro Recall

Recall measures the model's capability to recognize all relevant instances within a dataset

$$Recall_i = \frac{True\ Positives\ (TP)_i}{True\ Positives\ (TP)_i + False\ Negatives\ (FN)_i} \quad (7.1)$$

- True Positives (TP): The number of correctly predicted positive instances (instances correctly classified as belonging to the class of interest) for class i
- False Negatives (FN): The count of positive instances that are mistakenly predicted as negative, meaning the instances that are incorrectly classified as not belonging to the target class, for class i .

And Macro Recall is the averaged recall across all classes.

$$Macro Recall = \frac{1}{N} \sum_{i=1}^N Recall_i \quad (7.2)$$

Where:

- N is the number of classes.

7.2 Macro Precision

Precision quantifies how many of the instances predicted as positive are actually positive. It focuses on the correctness of the positive predictions made by the model. Precision is inversely related to the number of false positives. Increased precision signifies a reduction in the number of false positives among the positive predictions.

$$Precision_i = \frac{True\ Positives\ (TP)_i}{True\ Positives\ (TP)_i + False\ Positives\ (FP)_i} \quad (7.3)$$

Where:

- True Positives (TP): The number of correctly predicted positive instances (instances correctly classified as belonging to the class of interest) for class i .
- False Positives (FP): The count of negative instances that were mistakenly identified as positive, meaning the instances that were incorrectly classified as belonging to the target class when they actually do not belong to it, for class i .

And Macro Precision is the averaged precision across all classes.

$$Macro\ Precision = \frac{1}{N} \sum_{i=1}^N Precision_i \quad (7.4)$$

Where:

- N is the number of classes.

7.3 Macro Area Under the Curve (AUC)

AUC evaluates a model's ability to differentiate between positive and negative classes across all classifications. AUC is less sensitive to class imbalance and provides a thorough evaluation of a model's effectiveness to separate classes.

$$Recall_i = \frac{True\ Positives\ (TP)_i}{True\ Positives\ (TP)_i + False\ Negatives\ (FN)_i} \quad (7.1)$$

$$False\ Positive\ Rate\ (FPR)_i = \frac{False\ Positives\ (FP)_i}{False\ Positives\ (FP)_i + True\ Negatives\ (TN)_i} \quad (7.5)$$

- Where False Positive Rate (FPR)_i denotes the ratio of actual negative instances that have been erroneously classified as positive for class *i*.

$$AUC_i = \int_0^1 Recall_i(FPR)_i d(FPR)_i \quad (7.6)$$

For a multi-class classification, AUC extends to measure the model's performance across multiple classes. Using the One-vs-Rest approach, the problem is broken down into binary tasks, where each class is treated as positive against the rest. The overall AUC is then averaged across all classes. Given the scope of this investigation and the challenges with the datasets, AUC stands out as a highly suitable metric for capturing the model's performance. Expanding this for multi class classification:

$$Macro\ AUC = \frac{1}{C} \sum_{i=1}^C AUC_i \quad (7.7)$$

Where:

- C is the total number of classes.
- AUC_i is the AUC of the binary classifier for class *i*.

7.4 Macro F1

The F1 Score encompasses both the correctness of positive predictions (precision) and the effectiveness in identifying all positive instances (recall), ensuring that both metrics are considered equally important. In cases where one class is significantly more frequent than the other, precision and recall might be misleading if used in isolation.

$$F1\ Score_i = 2 \times \frac{Precision_i \times Recall_i}{Precision_i + Recall_i} \quad (7.8)$$

Where:

- Precision is the proportion of true positive predictions from all positive predictions made by the model for class *i*
- Recall is the proportion of true positive predictions from all actual positive instances in the data for class *i*

And Macro F1 is the averaged F1 scores across all classes.

$$\text{Macro F1 Score} = \frac{1}{N} \sum_{i=1}^N \text{F1 Score}_i \quad (7.9)$$

Where:

- N is the number of classes.

8. Novel Approach to Model

The common theme among the top three performing models in DFUC2021 was their ability to maintain a high resolution throughout the CNN network by utilizing deeper models with more parameters and then combining them into an ensemble averager. Due to computational restrains, an ensemble averager was not possible to implement. But HRNet-W32 - which has 25.5 million parameters - was chosen because it can preserve high-resolution representations throughout the network. This is achieved by creating a flow of information from high to low resolution through lateral connections and integrating parallel branches for extracting features at multiple scales. This approach ensures that vital high-resolution information is maintained and transmitted throughout the network, which is especially important when handling intricate visual inputs with fine details (Wang, J 2020).

Observations in the literature have identified a high correlation between generalisation error and the sharpness of the training loss, or how quickly it changes around the model's parameters (Keskar, 2016). Due to this, Sharpness-Aware Minimization (SAM) is an optimisation technique first presented by Foret et al. (2021). During training, optimization algorithms like stochastic gradient descent update model parameters (weights and biases) by computing gradients based on a randomly selected subset of the training data, which introduces variability that helps minimise the loss function (Medium, 2021). If the

loss landscape is "sharp," small changes in these parameters can lead to large increases in the loss function, which means the model's performance can degrade significantly with slight parameter perturbations. This suggests that the model might be overfitting the training data and struggle to generalise well to new data. SAM intervenes by guiding the optimisation process to prefer parameter configurations where the loss landscape is "flatter." In a flatter landscape, small changes in the parameters do not cause significant increases in the loss. This means the model is less sensitive to minor adjustments, leading to better generalization and robustness (Kim, 2023). As a result, a SAM wrapped SGD optimizer with a learning rate of $1e-3$ was integrated with the HRNet model.

SAM does this by first slightly perturbing the model's parameters in different directions to assess the sharpness of the loss function. It then adjusts the parameters to minimize the loss while considering these perturbations, pushing the model toward flatter, more stable minima. Following the evaluation of the loss at these perturbed points, SAM resets the parameters to their initial values and executes the normal SGD update based on the computed gradients, modifying them to minimise the loss while taking the perturbations into account. As a result, model training will take twice as long to run, but the significant improvement in model performance when using SAM wrapped around the SGD optimizer compared to standard SGD alone was highlighted in the winner of the DFUC2021 competition (Galdran, 2021). The complexities involved in classifying different DFU types suggest that, beyond a certain point, small parameter adjustments aimed at minimizing the loss function can become counterproductive as overly fine-tuning the model to reduce loss might lead to overfitting.

HRNet uses global pooling to collapse spatial dimensions to a 1-dimensional vector before reaching the penultimate layer, and the penultimate layer of all of the models employed during this investigation was modified to accommodate a classification head for four classes corresponding to the four DFU classifications. ResNext-50 and VGG-19 were also employed for comparative purposes to provide a more comprehensive examination of the performance of the modified HRNet architecture. All models were pre-trained on the ImageNet dataset - which has 1.2 million images across 1,000 categories in its training set - to leverage big image transfer. This provides a strong baseline for training by capturing general features learned from a large and diverse set of images.

A batch size of 32 was used, and all models were trained for 20 epochs, with the weights and biases from the best-performing epoch applied to the final test set. These decisions were made to maximise performance while accounting for computational constraints.

9. Results and Discussion

The following tables present the respective validation and test results for each model on the datasets curated during this investigation. Tables 3 and 4 display the results for ResNeXt-50, Tables 5 and 6 show the results for VGG-10, and Tables 7 and 8 provide the results for the modified HRNet-W32 architecture.

ResNeXt	Best Epoch	Validation AuC	Validation Recall	Validation Precision	Validation F1
DFUC2021	19	0.881	0.566	0.584	0.529
Aug-DFUC2021	20	0.895	0.585	0.605	0.595
Aug-DFUC2021 (100%)	19	0.886	0.575	0.594	0.582
Aug-DFUC2021 (95%)	17	0.875	0.590	0.555	0.559
Aug-DFUC2021 (90%)	20	0.878	0.790	0.553	0.535
Aug-DFUC2021 (85%)	20	0.801	0.590	0.525	0.505
Aug-DFUC2021 (80%)	18	0.777	0.524	0.473	0.485

Table 3 – ResNeXt-50 Macro Validation Performances on Respective Datasets.

VGG-19	Best Epoch	Validation AuC	Validation Recall	Validation Precision	Validation F1
DFUC2021	19	0.875	0.595	0.543	0.573
Aug-DFUC2021	20	0.902	0.628	0.663	0.655
Aug-DFUC2021 (100%)	17	0.895	0.621	0.658	0.644
Aug-DFUC2021 (95%)	18	0.882	0.609	0.648	0.628
Aug-DFUC2021 (90%)	20	0.840	0.590	0.567	0.569
Aug-DFUC2021 (85%)	18	0.821	0.530	0.490	0.513
Aug-DFUC2021	19	0.804	0.507	0.467	0.424

Table 5 - VGG-19 Macro Validation Performances on Respective Datasets.

ResNeXt	Test AuC	Test Recall	Test Precision	Test F1
DFUC2021	0.823	0.503	0.531	0.489
Aug-DFUC2021	0.802	0.510	0.453	0.418
Aug-DFUC2021 (100%)	0.804	0.532	0.475	0.433
Aug-DFUC2021 (95%)	0.823	0.494	0.510	0.491
Aug-DFUC2021 (90%)	0.839	0.490	0.456	0.478
Aug-DFUC2021 (85%)	0.763	0.494	0.400	0.402
Aug-DFUC2021 (80%)	0.735	0.491	0.398	0.382

Table 4 – ResNeXt-50 Macro Test Performances on Respective Datasets.

VGG19	Test AuC	Test Recall	Test Precision	Test F1
DFUC2021	0.835	0.512	0.587	0.480
Aug-DFUC2021	0.782	0.540	0.444	0.450
Aug-DFUC2021 (100%)	0.801	0.551	0.485	0.494
Aug-DFUC2021 (95%)	0.849	0.566	0.544	0.540
Aug-DFUC2021 (90%)	0.814	0.560	0.470	0.483
Aug-DFUC2021 (85%)	0.795	0.488	0.436	0.441
Aug-DFUC2021 (80%)	0.789	0.501	0.435	0.446

Table 6 - VGG-19 Macro Test Performances on Respective Datasets.

HRNet	Best Epoch	Validation AuC	Validation Recall	Validation Precision	Validation F1
DFUC2021	17	0.894	0.627	0.577	0.607
Aug-DFUC2021	19	0.933	0.659	0.693	0.676
Aug-DFUC2020 (100%)	20	0.927	0.653	0.688	0.668
Aug-DFUC2021 (95%)	16	0.918	0.634	0.681	0.657
Aug-DFUC2021 (90%)	19	0.871	0.617	0.593	0.605
Aug-DFUC2021 (85%)	20	0.784	0.558	0.527	0.533
Aug-DFUC2021 (80%)	18	0.627	0.412	0.459	0.439

Table 7 - Modified HRNet Macro Validation Performances on Respective Datasets.

HRNet	Test AuC	Test Recall	Test Precision	Test F1
DFUC2021	0.856	0.541	0.530	0.501
Aug-DFUC2021	0.866	0.580	0.529	0.493
Aug-DFUC2021 (100%)	0.847	0.567	0.520	0.520
Aug-DFUC2021 (95%)	0.879	0.600	0.593	0.587
Aug-DFUC2021 (90%)	0.863	0.572	0.531	0.546
Aug-DFUC2021 (85%)	0.742	0.501	0.413	0.372
Aug-DFUC2021 (80%)	0.573	0.269	0.313	0.278

Table 8 - Modified HRNet Macro Test Performances on Respective Datasets.

9.1 Overall Model Performance Evaluation

The investigation explores the performance of three deep learning models – a modified HRNet-W32, VGG-19, and ResNeXt-50 - on novel datasets derived from the DFUC2021 dataset. These datasets incorporated applied data augmentations and/or near-image-deduplication, and the model performances were evaluated on each - the original DFUC2021 dataset and various modified versions, with similarity thresholds set at 100%, 95%, 90%, 85%, and 80%, as well as on an augmented dataset that didn't undergo deduplication. The results revealed significant insights into the impact of augmentation and model architectures on classification. The modified HRNet exhibited the strongest test performance on the 95% similarity threshold dataset, achieving the highest macro F-1 score, precision, recall and AUC performance amongst all the models and datasets investigated. This indicates the effectiveness of the novel dataset creation strategy and the modified HRNet architecture.

However, as the similarity threshold decreased, HRNet's test performance deteriorated more sharply compared to other models. This was particularly evident in its performance on the 80% similarity threshold dataset, where it significantly underperformed in comparison to VGG-19 and ResNeXt-50 (0.278 Macro F-1 vs 0.446 and 0.382 for VGG-19 and ResNeXt-50 respectively). This decline can be attributed to HRNet-W32's architecture, which is designed to maintain high spatial resolution throughout its layers, which is ideal for tasks requiring detailed feature extraction such as DFU classification. When the dataset is heavily pruned, as in the case of the 80% threshold, the reduced number of unique training samples limits the model's ability to capture fine-grained details, leading to poorer performance. Higher thresholds (95% and 100%) retain more unique and slightly varied images, which better suit HRNet's ability to learn subtle spatial differences.

9.2 Augmentation Evaluation

The comparison of both test and validation scores between the original DFUC2021 dataset and the augmented dataset (Aug-DFUC2021) that did not undergo deduplication for all three models investigated provides a gauge of the effectiveness of the augmentation techniques. A clear trend emerged: models trained on the original dataset generally outperformed those trained on the augmented, balanced dataset (Aug-DFUC2021) in test performance, while the opposite was true for validation performance.

Although the primary goal of augmentation is to introduce variability that mimics real-world scenarios and helps models generalize better, inappropriate or excessive augmentation can lead to overfitting. This issue was particularly pronounced for classes the ischaemia and both, where augmentation increased the number of samples by approximately 11.4-fold and 4.1-fold, respectively. Simple extrapolation dictates that, on average, each original image present in ischaemia and both classes were augmented around 11.4 or 4.1 times, producing numerous variations of the same image. Multiple versions of the same underlying patterns flood the model training process, especially when the augmentations fail to simulate new, realistic variations. As a result, the model learns the specific patterns of the data instead of more generalisable features and becomes more prone to overfitting when faced with real-world data. While SAM was introduced to negate overfitting, its effectiveness was undermined by the limitations in the quality of the augmented dataset and significant overfitting was observed for models when utilising this dataset.

The majority classes in the original dataset (infection and none) contain a wider variety of samples, offering the model a richer set of features to learn from. Aggressive dataset balancing through augmentation significantly diluted the natural variation in the minority classes, introducing noise and distorting critical features necessary for accurate classification. Since the validation set was derived from 10% of the training data, it could have potentially included multiple near-duplicated images.

Diabetic foot ulcers are multifactorial and incredibly nuanced, with successful classification hinging on the ability to differentiate between subtle fine-grained details such as colour, shape, and ulcer depth. If the augmentations produce misleading samples, i.e, where brightness is adjusted, this difference may cause the model to incorrectly distinguish between healthy and ulcerated tissue. In the fully augmented dataset, the presence of duplicated images in the validation set likely inflated performance metrics, suggesting that the data augmentations alone did not, overall, introduce meaningful variation.

9.3 Deduplication Evaluation

Across all models, a general trend was observed from 95% to 80% similarity thresholds: models utilising datasets with higher thresholds consistently outperformed those that used lower. At the 95% threshold, the process effectively identified and eliminated identical or near-duplicate images that could hinder model performance (as seen in the fully augmented dataset). At the same time, it retained enough training samples with sufficient diversity to enhance model performance compared to the original dataset, with all models achieving superior macro f-1 scores when utilising Aug-DFUC2021 (95%) vs the original DFUC2021 dataset. This suggests that for this similarity threshold an optimal balance was achieved between removing duplicate images, sample variation, maintaining an adequate sample size, and equal class sample distribution.

The 80% similarity threshold exhibited notably poor performance throughout the investigation, which can be attributed to the dataset containing only 2,035 samples. The dilemma of limited training data and its impact on model performance had already been identified when researching the literature, and while the 80% threshold resulted in a more evenly distributed class distribution, the drastic reduction in dataset size to roughly one-third of the original outweighed the supposed benefits of class balance.

Further investigation into the deduplication revealed false positives for images identified as duplicates, with the misclassification becoming more prevalent as the similarity threshold decreased. Figures 13 and 14 provide examples of images that were incorrectly classified as near duplicates.



Figure 13 – Misclassified as a duplicate image, Classification None



Figure 14– Misclassified as a duplicate image, Classification Ischaemia

This issue arises from the limitations of hash functions used to identify duplicates. Although hash functions can generate a vast number of possible combinations, there is always a probability that completely different images may produce the same or very similar hash - a phenomenon known as a hash collision (Cherckesova, 2022). While using larger hash sizes helps mitigate this risk, it cannot be completely eliminated. Coinciding with the increasing number of images in a dataset, the probability of hash collisions also rises, a concept related to the birthday paradox (Zilles, 2007), where the likelihood of two distinct images producing the same hash rises with the number of comparisons. This effect becomes more pronounced at lower similarity thresholds, where the deduplication process mistakenly removes non-duplicate/non-near duplicate images due to the similarity of the 256-bit string, further reducing dataset size and diversity. Thus, while applying a similarity threshold can improve model performance up to a certain point (e.g., 95%), lowering the threshold beyond this point, actually

diminishes model performance by excessively reducing the dataset and introducing errors in near-duplicate image detection.

10. Future Work

To enhance the training process and model performance, future research should be conducted within environments with greater computational resources. This investigation was performed using Google Colab Pro and while able to make use of virtual GPU's, the limited runtimes and restricted number of compute units, available only on a pay as you go basis, proved to be a significant obstacle during this investigation. By being able to conduct a grid search to determine optimum hyperparameters (batch size, number of epochs, learning rate ect.), models can be trained more effectively, leading to increased performance. This investigation utilised a modified HRNet-W32 with 25.5 million parameters. As identified previously, deeper models typically outperform shallower ones for the task of DFU classification. As a result, enhanced computational power would allow for deeper models to be investigated (i.e. HRNet-W48 with 48 million parameters) and incorporate ensemble averagers, which constituted the top three performing models in DFUC2021. Systematically exploring various ensemble combinations and their impact on performance could yield valuable insights into optimising DFU image classification tasks.

Increasing the sensitivity of the deduplication process could improve on the research outcomes of this investigation. Narrowing the range of similarity thresholds while increasing the increments would provide a more refined deduplication process. The models trained on the Aug-DFUC2021(95%) dataset outperformed those using the original DFUC2021 data. Interestingly, models using Aug-DFUC2021 (90%) yielded results that were comparable to those of the original DFUC2021 dataset. Both datasets contain a similar number of training samples (6705 vs 5955), and despite the expectation that DFUC2021 (90%) would have superior quality due to undergoing near duplicate image identification and removal, this was not consistently reflected within the results. In order to further assess dataset quality, performing the same investigation using datasets with 1% increments between 100% and 90% similarity thresholds would provide a more comprehensive indication of both dataset quality and augmentation quality.

Another possible avenue for future work involves making use of unlabelled samples through semi-supervised learning. Bloch et al. (2021), who placed second place in DFUC2021, implemented semi-supervised learning during their research. By integrating unlabelled data into the training process, models can leverage additional information to improve performance. Considering the problem of insufficient training samples, making use of these samples would provide greater variability of images corresponding to their respective classifications, and should theoretically improving overall model ability in generalising.

A fully comprehensive numerical study of the hash set algorithms implemented for this research would further build on the findings presented in this report. Examining the classification of near-duplicate images using quantitative measures can help clarify the relative advantages and disadvantages of the various hashing algorithm. This analysis could inform the selection and refinement of hashing methods to enhance overall dataset quality and resultant model performance.

11. Conclusion

The potential efficacy of the remote classification of diabetic foot ulcers based on images alone has caused a surge in research surrounding the topic area in the literature. This thesis identified issues related to the quality of the DFUC2021 dataset with respect to the presence of near duplicate images, proven true by the improved performance when comparing models run using the original dataset DFUC2021 with novel DFUC2021 (95%), where images classified as being more than 95% similar were removed. Excessive augmentation led to an increase in duplicates which negatively impacted model performance as evidenced by the decreased macro AUC and F1 scores in the augmented dataset with no similarity threshold (AUG-DFUC). Using 90% and 95% similarity thresholds mitigated these issues resulting in better performance; however, due to the inherent limitation of hashing algorithms, decreasing similarity thresholds beyond this point led to performance degradation. This was caused by insufficient sample sizes as near-duplicate images were misclassified, a problem linked to the birthday paradox.

The findings emphasize the importance of balancing augmentation strategies and carefully selecting complimentary hashing techniques to optimize performance in image classification. Furthermore, they highlight the need for a larger, more diverse dataset for DFU classification in order to further improve on deep learning performance with the ultimate aim of entirely replacing the current diagnostic methods for diabetic foot ulcers.

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Appendix A – Terms of Reference (TOR)

7Z10SS Masters Project

NPC

ToR Coversheet

Department of Computing and Mathematics Computing and Digital Technology Postgraduate Programmes Terms of Reference Coversheet	
Student name:	Omari Ricketts
University I.D.:	21411371
Academic supervisor:	Moi Hoon Yap
External collaborator (optional):	
Project title:	Analysis of the Diabetic Foot Ulcer Image Datasets
Degree title:	MSc Data Science
Project unit code:	6G7V0007
Credit rating:	60
Start date:	18/06/2024
ToR date:	21/06/2024
Intended submission date:	27/09/2024
Signature and date student:	Omari Ricketts 18/06/2024
Signature and date external collaborator (if involved):	

This sheet should be attached to the front of the completed ToR and uploaded with it to Moodle.

Project Title

Analysis of the Diabetic Foot Ulcer Image Datasets

Project Background

Images of DFUs play a crucial role in visually classifying, diagnosing and monitoring the condition, allowing healthcare professionals to accurately assess the severity and progression of ulcers (Cassidy, 2021b) (Goyal, 2018b). This visual information is indispensable for developing tailored treatment plans and intervening promptly to prevent complications such as infections, gangrene, or even the need for amputations. However, where present, these factors introduce variability that makes it challenging to classify ulcers based on visual inspection alone.

Diabetic foot ulcer datasets serve as crucial resources for researchers developing new computer algorithms for DFU analysis. Over the past two years, there has been a concentrated effort on detecting and classifying these ulcers (Cassidy, 2021b) (Yap, 2021a). (Yap, 2021b) (Goyal, 2018b). These datasets include ulcer or wound photographs accompanied by expert-provided ground-truth labels. However, the data quality significantly impacts the outcomes and reliability of subsequent analyses.

Issues such as poor photo quality, inadequate lighting, and insufficient training data hinder the effectiveness of deep learning methods for DFU classification. Additionally, the presence of duplicate images—whether from including the same ulcer or wound photograph multiple times or artificially augmenting the dataset to overcome the lack of adequate data—can distort algorithmic results. This redundancy may lead to overfitting, where models learn to recognize specific images rather than general ulcer features (Manjusha, 2022). To maintain the integrity and utility of the DFU datasets, it is crucial to implement stringent data curation processes that identify and remove duplicates, ensuring that each image contributes uniquely to the research objectives (Manjusha, 2022). Addressing this issue is essential for fostering meaningful advancements in DFU detection and classification, facilitating more reliable and impactful research outcomes.

The data that will be used for the investigation was provided by Lancashire Teaching Hospitals and is distributed across two distinct datasets. The “DFUC2021” dataset contains 9949 samples for training, of which 3994 are unlabeled DFU patches, making this a supervised learning task. There are 5,955 samples designated for testing, and since the data is pre-divided into training and testing, only the validation set needs to be generated from the training data. The ground truth labels for this dataset fall into four classes: control, infection, ischaemia and both conditions.

The other dataset is divided into two parts and includes samples classified into six categories: normal (642 samples), abnormal (1,039 samples), infection (positive or negative, with 2,945 samples each), and ischaemia (positive or negative, with 4,935 samples each). Manual creation of the training, testing, and validation sets will be necessary, and the datasets will be curated to ensure alignment of classification.

Aim

The aim of this research is to develop an automated computer method for diabetic foot ulcer image classification and assess the performance on a curated dataset. This will establish a new benchmark by identifying, analysing, and appropriately removing duplicates within and across DFU datasets, particularly across testing and training sets.

Objectives

In order to achieve the aims identified various tasks need to be completed:

- Conduct a literature review to gain a comprehensive understanding of existing research and methodologies for detecting and handling duplicate images in medical datasets, as well as the architectures currently used for classifying diabetic foot ulcers.
- Develop a modified deep learning architecture for classifying diabetic foot ulcers.
- Analyse the performance of common deep learning architectures on the new, curated dataset
- Conduct a critical performance analysis of the modified architecture developed in this investigation, comparing it with commonly used architectures from the literature on the curated dataset

Evaluation plan

Regarding data curation, the impact of removing duplicates will be assessed based on the number of duplicates successfully identified and removed, as well as the resulting dataset size. Evaluating these factors will highlight the effectiveness of the curation process in enhancing dataset integrity.

Additionally, the integrity of the dataset will be evaluated by comparing model performance before and after curation to determine any improvements or changes.

The results from the classification element of the investigation will be evaluated using four metrics. Cross-entropy loss, which quantifies the divergence in probability between the predicted and actual classifications. Precision assesses the accuracy of positive predictions, indicating the model's capability to correctly identify relevant instances among all predicted positives. Recall measures the proportion of actual positives correctly identified by the model, indicating its sensitivity to detecting true positives, and the F1 score, which balances precision and recall, provides a comprehensive evaluation of the model's overall performance by accounting for both false positives and false negatives (Yap, 2021a)

Appendix B – Activity Schedule

The project timeline was structured using a Gantt chart, which detailed the start dates and deadlines for each phase of the project

Task	Start Date	End Date	Duration (Days)
ToR and Ethics	01/06/2024	21/06/2024	20
Literature Review	01/06/2024	08/07/2024	37
Model Building and Data Curation	09/07/2024	09/08/2024	31
Collation of Results	10/08/2024	17/08/2024	7
Report Writing	18/08/2024	27/09/2024	40

